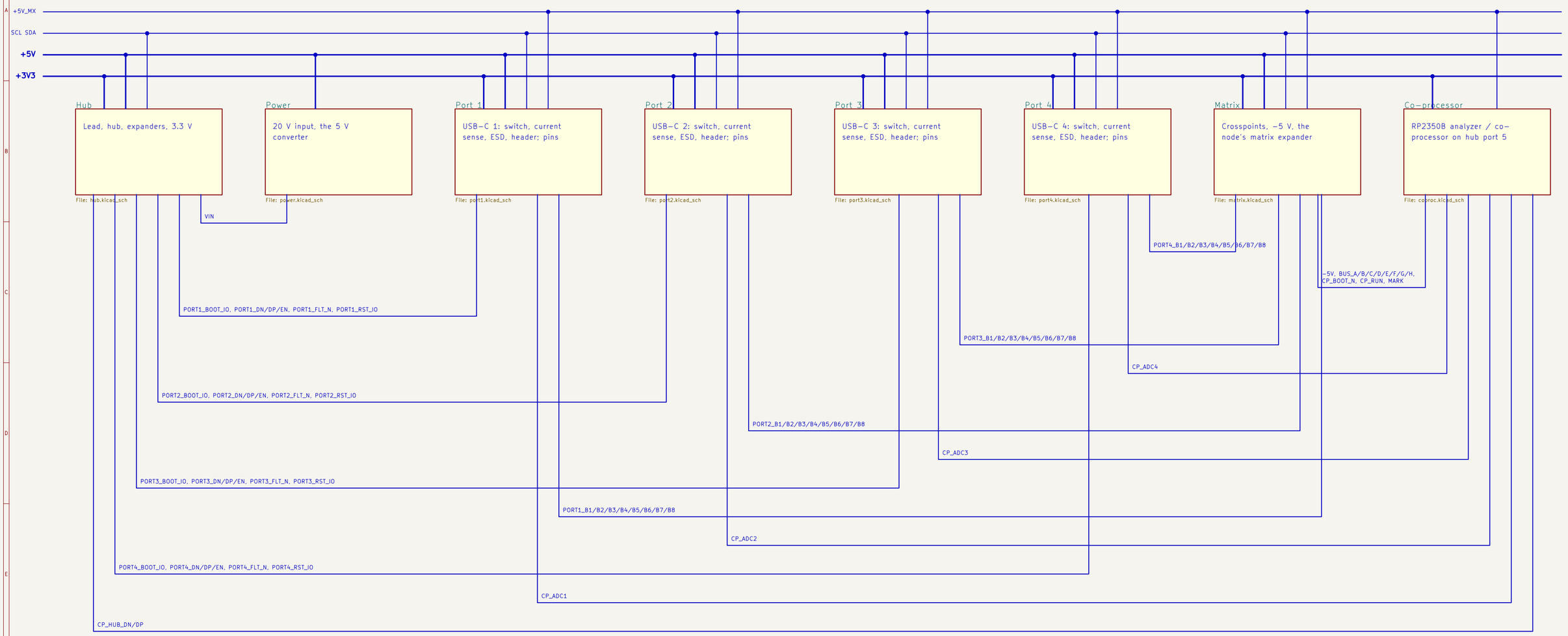


DATA ×4 v1.0 — USB2517I hub (7 ports), four USB-C (sub-slots A1.1-A1.4) each with a TPS2553-1, an INA226 and ESD, two TCA6408A for EN/FAULT/RESET/BOOT, an LMR51635 5 V from the lead's 20 V, an AP2114H 3.3 V; the PH 8 lead from the node's J7.
Pins: 32 on two CH446Q (±5 V), LSF0108 per port, an RP2350B analyzer on hub port 5, a TCA9534A for the matrix and the co-processor's RUN/BOOTSEL. Standing behind the port wall, one face assembled; v1 stays the other bays' board.
Lead pinout here: 1 NC · 2 SCL · 3 SDA · 4 GND · 5 D− · 6 D+ · 7 GND · 8 VIN — the DATA ×6's lead.
Rails: VIN 20 V (on above 13.5 V typ., 15.1 V at worst; ≤ 28 V), ≤ 0.92 A at 20 V (1.22 A at 15 V) · +5V 5.08 V (4.94-5.22 V), ≤ 3.09 A (the ports at their 0.644 A limit + 0.51 A; the LMR51635's 3.5 A) · PORTn_VBUS 4.82-5.22 V, 0.52-0.64 A limit · +3V3 3.3 V (AP2114H), ≤ 0.51 A · +5V_MX, −5V (sheet Matrix) · +3V3_CP, +1V1 (sheet Co-processor) · HUB_1V8 / HUB_1V8PLL 1.8 V (the hub's own regulators).
I²C: TCA6408A 0x20 / 0x21, INA226 0x41 / 0x43 / 0x44 / 0x45, TCA9534A 0x38; the node's INA226 0x40.
Assembled as DATA-4x-logic (as drawn) or DATA-4x (BOM/CPL '-base': the matrix, the co-processor and the ports' pin chains not fitted).

Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack

Sheet: /
File: data-port-x4.kicad_sch

Title: DATA ×4 v2 — Overview

Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/9

Lead from the bay node's J7 (JST PH 8, vertical, on the rear face): SDA and SCL through a H5VU25UC; pin 8 VIN (sheet F2.4682 A) to 5 V

3.3 V for the hub, the expanders and the INA226s (AP2114H, SOT-223 on copper)

Hub: CFG_SEL[2:0] = 110 (internal default configuration, straps disabled), 24 MHz crystal

Two TCA6408A expanders, one per pair of ports: EN, FAULT_N, RESET, BOOT

Bring-up test points on the rear face (B.Cu, bare pads, not in the BOM): +3V3, +5V, the hub's reset and 1.8 V, the lead's I²C, the expanders' RESET, GND

I²C isolation: open with VIN off, so the node's pull-ups cannot feed this board

Hub reset: U5 (MAX809S, 2.93 V, ≥ 150 ms, push-pull) on RESET_N (errata DS80000628A, Module 2).
Hub: CFG_SEL[2:0] = 110 (internal default configuration, strap options disabled: DS00001598C Table 5-2):
  self-powered, ports switched and sensed one by one (here by the expanders); no port disabled. D+ / D− not
  sampled at reset; the other strap pins keep their pull-downs, unread. Ports 6 and 7 enabled, unused
  (cosmetic). On purpose: PRTPWR unused and OCS_N on their pull-ups (the node switches each port and reads
  its FAULT_N through the expanders); VBUS_DET on 3.3 V (self-powered hub).
Hub port n = USB-C n (build 1–14) for n = 1–4; port 5 = the co-processor.
Expander pins are inputs at power-up: the EN pull-downs keep every port off until the node writes.
No INT on the lead (J1.1 open); the node polls the expanders and the INA226s every 10–20 ms.
I²C: the node's pull-ups on LEAD_*. 4.7k / 4.7k here; with VIN off the chips here lose their registers:
  the node sets them up again after each VIN on, and leaves this board's addresses alone while off.
J1 (through-hole) stands on the rear face; soldered by hand (the PCBA places the front face only).

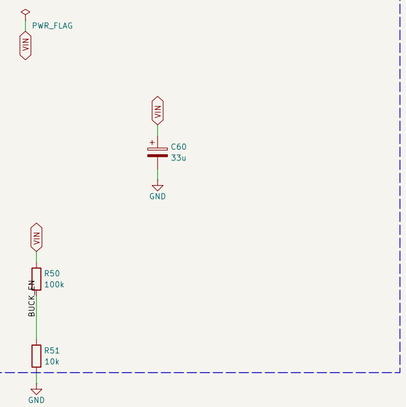
Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, 8 logic an
hw-testing-rack

Sheet: /Hub/
File: hub.kicad_sch

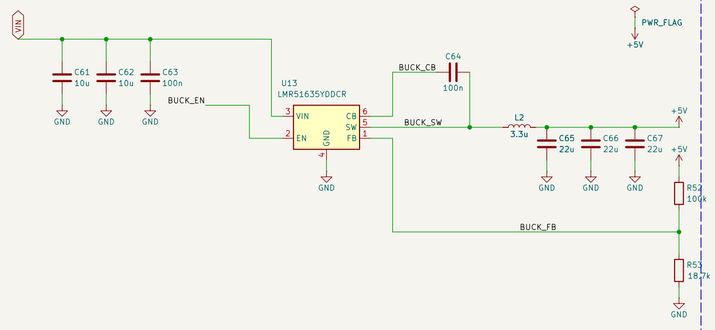
Title: DATA x4 v2 — Lead, hub, expanders, 3.3 V

Size: A1 | Date: | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 2/9

VIN: the rack's 20 V, switched by the node (its eFuse), from the PH-8 lead's pin 8 (J1, sheet Hub)



+5V: LMR51635, synchronous, 1.1 MHz, 3.5 A (the DATA ×6's block, TI SLUSF64B table 8-2)



Bring-up test point on the rear face (B.Cu): VIN



The DATA ×6's converter with the family's 3.5 A part (TI SLUSF64B table 8-2, LMR51635, 5 V, 1.1 MHz: 3.3 µH
  Vishay IHLP2525CZ, 75 V, 13.5 A sat; 2 × 10 µF + 100 nF in, 3 × 22 µF out), PFM and spread spectrum as the
  ×6's LMR51625.
+5V: the four ports, the 3.3 V LDO (≤ 0.51 A) and the matrix: 2.51 A at 0.5 A a port; at worst, all four at
  the TPS2553s' 0.644 A limit, 3.09 A, under the LMR51635's 3.5 A with no help from the node's firmware.
  The inductor then peaks at 3.68 A from 28 V (1.17 A ripple), 3.98 A at the worst corner (L −20 %, 0.99 MHz), under the LMR51635's 4.2 A minimum peak limit (SLUSF64B §6.5: 4.2 / 5.1 / 6.0 A: valley 3.2 / 4.0 / 4.8 A);
  from 13.5 V: peak 3.53 A, valley 2.65 A. No frequency foldback (tON 170 ns ≥ 70, tOFF 556 ns ≥ 200).
FB 100k/18.7k: 5.08 V. 4.94-5.22 V with VFB 0.792-0.808 V (SLUSF64B §6.5) and the 1 % resistors: under
  the TPS60400's 5.25 V (U9, sheet Matrix), which the node's 18k (5.10-5.39 V) was not.
EN 100k/10k: on above 13.5 V typ., up to 14.8 V on the EN threshold's 1.103-1.349 V (SLUSF64B §6.5) and
  15.1 V with the 1 % resistors: the rack's 20 V clears it. Off below 10.8 V typ.
VIN on J1.8, the node's VRACK through its eFuse (TPS26600, off by default): no second protection here.
  At the 3.09 A worst case from 5.22 V (η ≈ 88 %): 0.92 A at 20 V, 1.22 A at 15 V, 1.36 A at 13.5 V, 0.65 A
  at 28 V; the converter's full 3.5 A: 1.38 A at 15 V. Under the node eFuse's 1.76 A: a PH contact carries 2 A.

Fiducials (top face, for the pick-and-place camera)



Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

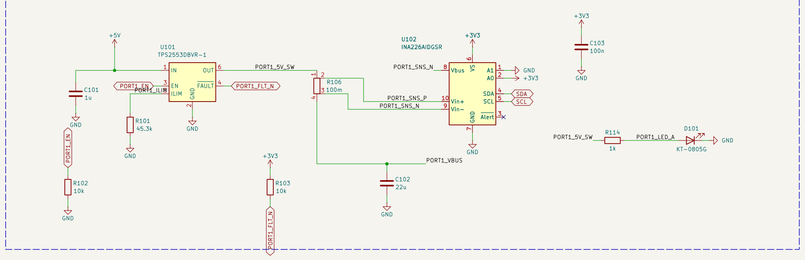
hw-testing-rack

Sheet: /Power/
File: power.kicad_sch

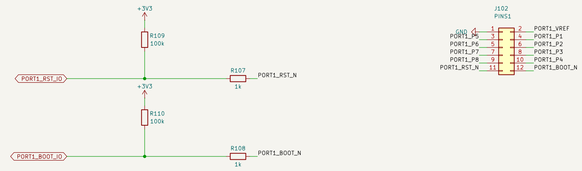
Title: DATA ×4 v2 — 20 V input, the 5 V converter

Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 3/9

Port 1 power: +5V → TPS2553-1 → PORT1_5V_SW → 100 mΩ shunt → PORT1_VBUS, the INA226 across the shunt



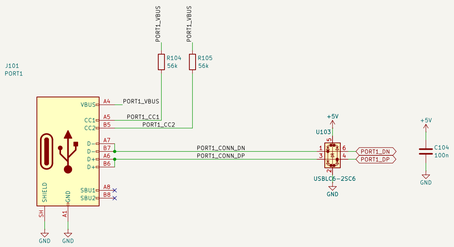
Header 1 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up



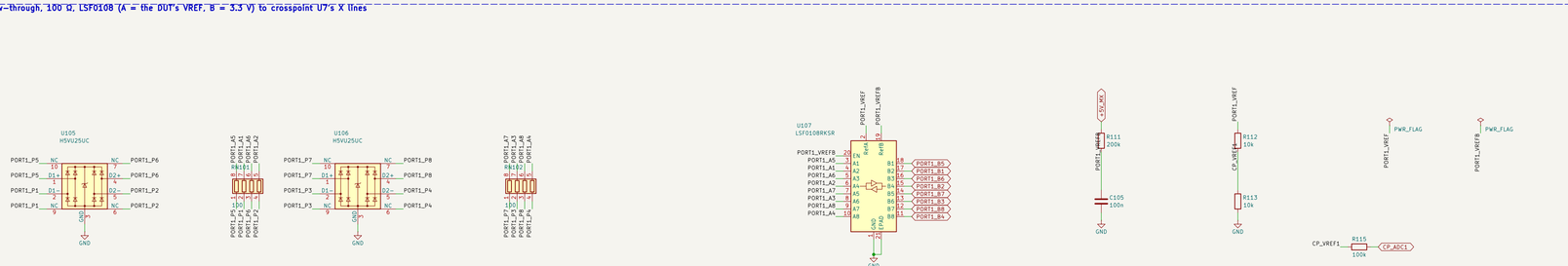
Bring-up test points on the rear face (B.Cu): FAULT_N, VBUS



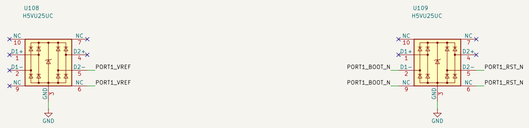
USB-C 1 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Pins 1: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines



ESD on the header's VREF (U108, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U109: header → ESD → 1k)

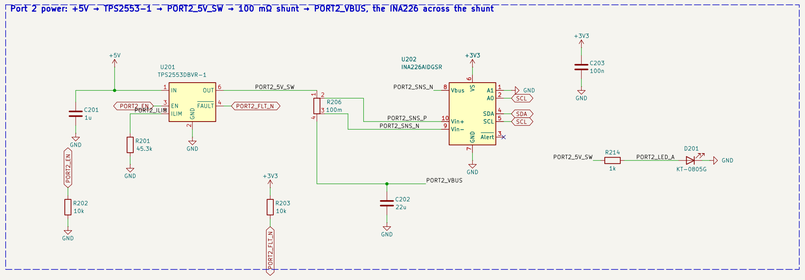


Port 1 = sub-slot A1.1, hub port 1, busid 1-1.1.
INA226 at 0x41: expander U2 (0x20).
ILIM 45.3k: 524 / 577 / 638 mA min / typ / max (TI SLVSB41G §8.5.1 eq. 1-3). 519-644 mA with the resistor's 1 %; the ~1 latches off after 7.5 ms in limit. FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA x4 v2 map):
an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U108, RESET_N / BOOT_N through U109 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V: a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS): a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

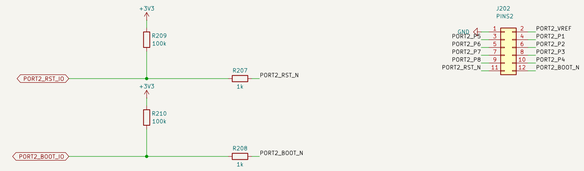
Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, 8 logic an

hw-testing-rack	
Sheet: /Port 1/	
File: port1.kicad_sch	
Title: DATA x4 v2 — USB-C 1: switch, current sense, ESD, header; pins	
Size: A1	Date:
KiCad E.D.A. 10.0.6	**Rev: 1.0**
	Id: 4/9

Port 2 power: +5V → TPS2553-1 → PORT2_5V_SW → 100 mΩ shunt → PORT2_VBUS, the INA226 across the shunt



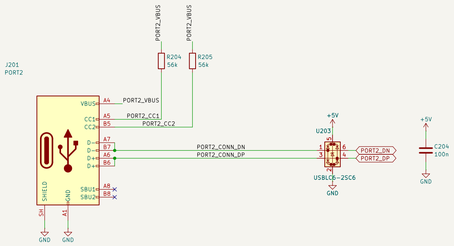
Header 2 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up



Bring-up test points on the rear face (B.Cu): FAULT_N, VBUS



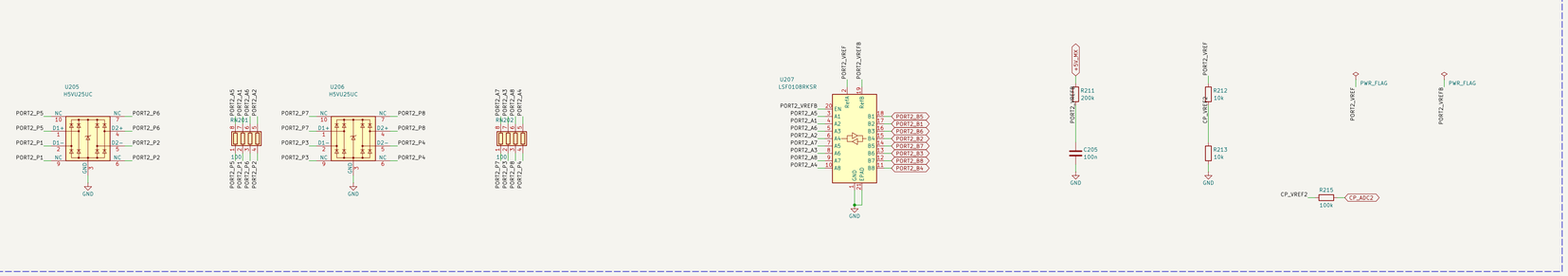
USB-C 2 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



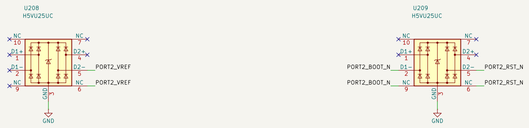
Notes

Port 2 = sub-slot A1.2, hub port 2, busid 1-1.2.
INA226 at 0x43: expander U2 (0x20).
ILIM 45.3k: 524 / 577 / 638 mA min / typ / max (TI SLVSB41G §8.5.1 eq. 1-3). 519-644 mA with the resistor's 1 %; the ~1 latches off after 7.5 ms in limit. FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 = crosspoint X lines (docs/electronics.md › DATA x4 v2 map):
an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U208, RESET_N / BOOT_N through U209 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V: a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS): a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

Pins 2: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines



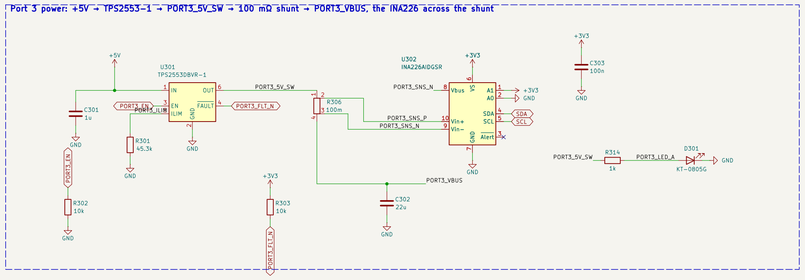
ESD on the header's VREF (U208, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U209: header → ESD → 1k)



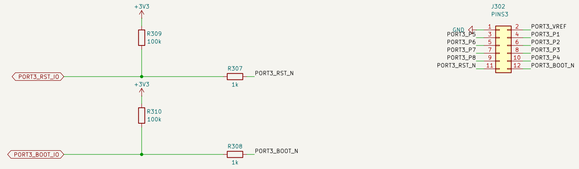
Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, 8 logic an

hw-testing-rack	
Sheet: /Port 2/	
File: port2.kicad_sch	
Title: DATA x4 v2 — USB-C 2: switch, current sense, ESD, header; pins	
Size: A1	Date:
KiCad E.D.A. 10.0.6	**Rev: 1.0**
	Id: 5/9

Port 3 power: +5V → TPS2553-1 → PORT3_5V_SW → 100 mΩ shunt → PORT3_VBUS, the INA226 across the shunt



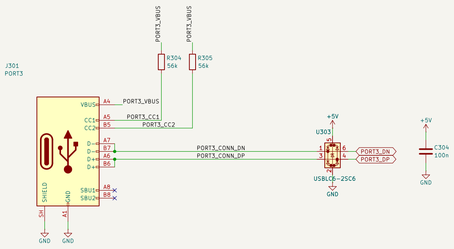
Header 3 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up



Bring-up test points on the rear face (B.Cu): FAULT_N, VBUS

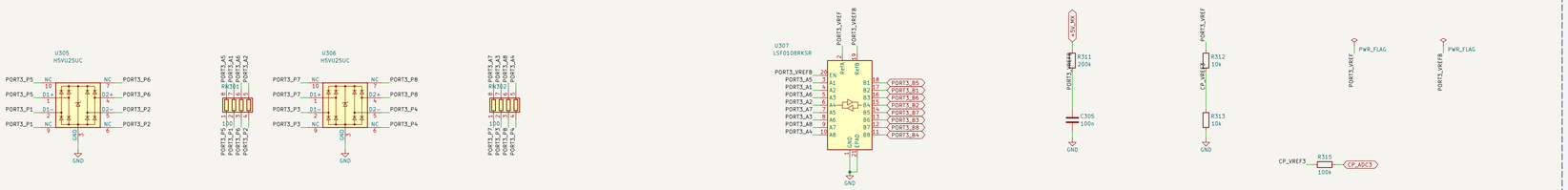


USB-C 3 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)

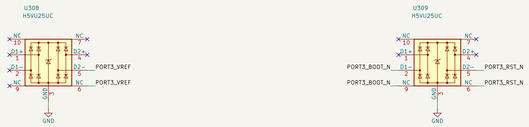


Port 3 = sub-slot A1.3, hub port 3, busid 1-1.3.
INA226 at 0x44: expander U6 (0x21).
ILIM 45.3k: 524 / 577 / 638 mA min / typ / max (TI SLVSB41G §8.5.1 eq. 1-3). 519-644 mA with the
  resistor's 1 %; the ~1 latches off after 7.5 ms in limit. FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA x4 v2 map):
  an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's
  crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U308, RESET_N / BOOT_N through U309 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V: a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS): a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

Pins 3: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines



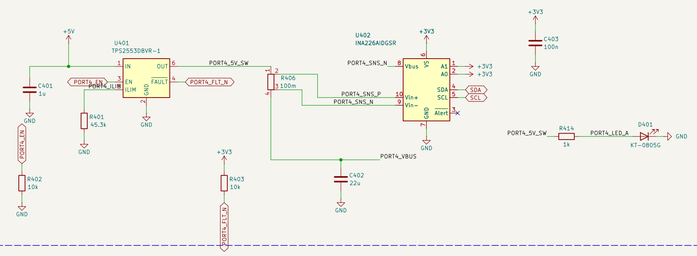
ESD on the header's VREF (U308, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U309: header → ESD → 1k)



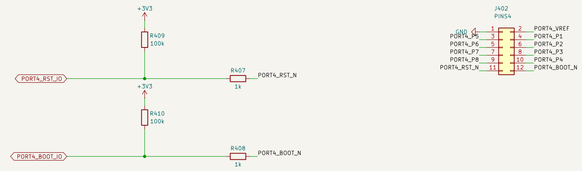
Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, 8 logic an

hw-testing-rack	
Sheet: /Port 3/	
File: port3.kicad_sch	
Title: DATA x4 v2 — USB-C 3: switch, current sense, ESD, header; pins	
Size: A1	Date:
KiCad E.D.A. 10.0.6	**Rev: 1.0**
	Id: 6/9

Port 4 power: +5V → TPS2553-1 → PORT4_5V_SW → 100 mΩ shunt → PORT4_VBUS, the INA226 across the shunt



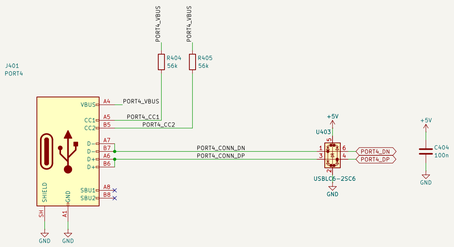
Header 4 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up



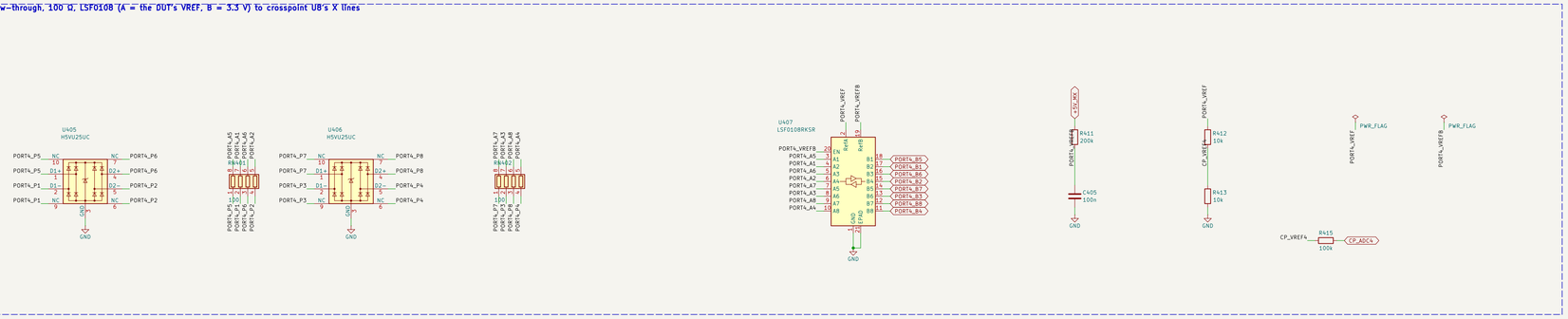
Bring-up test points on the rear face (B.Cu): FAULT_N, VBUS



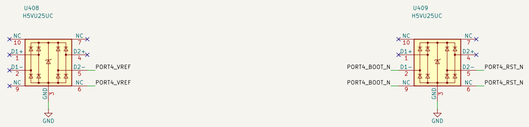
USB-C 4 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Pins 4: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines



ESD on the header's VREF (U408, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U409: header → ESD → 1k)

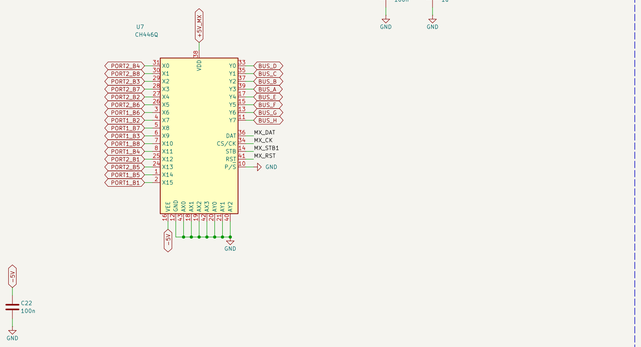


Port 4 = sub-slot A1.4, hub port 4, busid 1-1.4.
INA226 at 0x45: expander U6 (0x21).
ILIM 45.3k: 524 / 577 / 638 mA min / typ / max (TI SLVSB41G §8.5.1 eq. 1-3). 519-644 mA with the resistor's 1 %; the ~1 latches off after 7.5 ms in limit. FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA x4 v2 map):
an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U408, RESET_N / BOOT_N through U409 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V: a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS): a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, 8 logic an

hw-testing-rack	
Sheet: /Port 4/	
File: port4.kicad_sch	
Title: DATA x4 v2 — USB-C 4: switch, current sense, ESD, header; pins	
Size: A1	Date:
KiCad E.D.A. 10.0.6	**Rev: 1.0**
	Id: 7/9

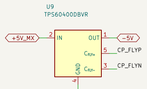
Crosspoint U7 (CH446Q at ±5 V, serial addressing): X = ports 1-2's pins, Y0-Y7 = buses DCBAEFGH





+5V_MX through a ferrite (the crosspoints, the inverter, the LSF0108s' bias); −5 V for the crosspoints' VEE (TPS60400 inverter)





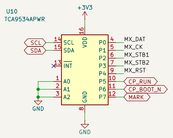






The node's matrix expander (TCA9534A, 0x38): the crosspoints and the co-processor's RUN / BOOTSEL

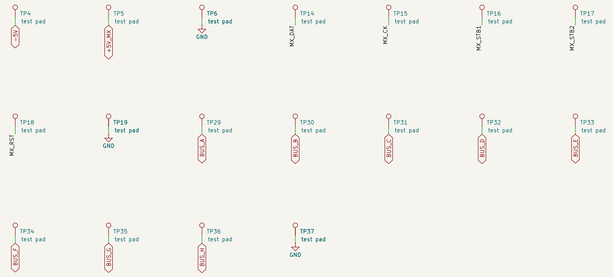






CH446Q serial mode (P/S low): 7 address bits on CK, data on DAT, latched by STB; RST high = all open.
TCA9534A 0x38: P0 DAT, P1 CK, P2 STB U7, P3 STB U8, P4 RST, P5 RUN, P6 BOOTSEL, P7 mark.
RUN and BOOTSEL are open-drain emulated (output register 0, the pin switched to output).
±5 V on purpose: at a single 5 V R_on is 120-225 Ω, and a 1.8 V board's I²C low would fail.

Bring-up test points on the rear face (B.Cu): ±5 V, the expander's bus and RST, buses A-H, GND (logic only)



Crosspoint U8 (CH446Q at ±5 V, serial addressing): X = ports 3-4's pins, Y0-Y7 = buses DCBAEFGH







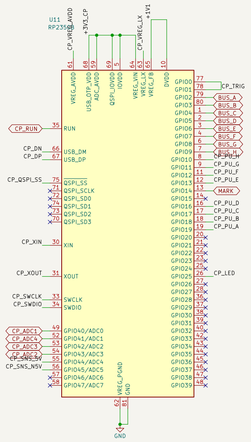
Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack
Sheet: /Matrix/
File: matrix.kicad_sch

Title: DATA ×4 v2 — Crosspoints, −5 V, the node's matrix expander

Size: A1	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 8/9

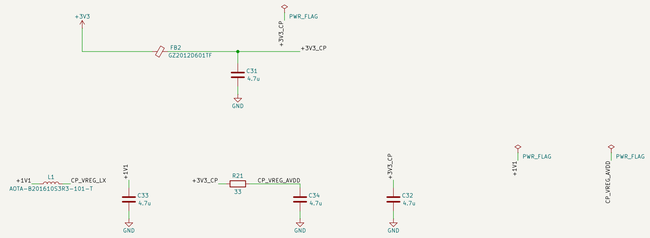
RP2350B, wired like a Pico 2 less its flash (the analyzer loaded into SRAM at power-up); its decoupling



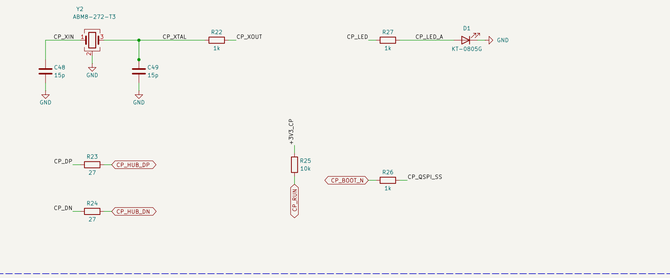


- GPIO2-9 = buses A-H (gusmanb CH1-8, 8-bit mode): GPIO0 → GPIO1 bridged (its trigger path).
- GPIO14 = the node's mark (TCA9534A P7); GPIO25 = LED; GPIO19…16, 13…10 = pull-up enables A…H (2.2k);
- GPIO40, 43, 42, 41 = VREF 1-4 / 2 (through 100k); GPIO44 = +5V × 47/147;
- GPIO45 = 3.3 V − (3.3 + 5) × 10/110 from −5 V (2.55 V nominal).
- RUN pulled up, BOOTSEL (QSPI_SS) through 1k: both held by the node (TCA9534A). open drain.
- Core regulator: Raspberry Pi's Minimal R4 block turned 180°: L1's dot (pad 1) on +1V1, not VREG_LX.
- C47 (DVDD pin 32, furthest from the regulator) 4.7 µF: no copper under LX / L1 (In1, In2, B.Cu).

Its 3.3 V from +3V3 through a ferrite, the core regulator as Raspberry Pi lays it out (3.3 µH, 3 × 4.7 µF, 33 Ω)



12 MHz crystal (ABM8-272-T3, 15 pF, 1k on XOUT), USB on hub port 5 (27 Ω), RUN, BOOTSEL, LED, SWD



Bus pull-ups: 10k to 3.3 V always (the LSFs' B side), 2.2k switched by the co-processor (GPIO10-13, 16-19); the ±5 V rails into the ADC



Bring-up test points on the rear face (B.Cu): SWD, RUN, BOOTSEL, its rails, GND (logic only)



SWD header (J2, 1 × 5 through-hole, fitted by hand): the Debug Probe on 1-3, RUN and VTref for others



Generated from electronics/data-port-v4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

hw-testing-rack

Sheet: /Co-processor/
File: coproc.kicad_sch

Title: DATA ×4 v2 — RP2350B analyzer / co-processor on hub port 5

Size: A1	Date:	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 9/9